INDEPENDENT POST-COURSE RESEARCH EXTENSION

Representation Drift Lab

When a model learns something new, what does it forget - and can changes inside the model warn us before capabilities are lost?

MEASURED METHODS	DOMAIN SCENARIOS	AUTOMATED CHECKS
9	**3**	**39**
Each run across 3 independent seeds	Food, satellite, pet, object, and digit data	Research, artifact, and web tests

A portfolio research project by **Ali Hasan**
Original 2025 course project with Sahil Mhatre and Corey Chen
Independent extension: Ali Hasan, 2026

Evidence status: LOCAL MULTI-SEED PRELIMINARY. The system and methods are real; the local samples are intentionally small and do not establish universal model rankings.

Artifact ID: clip-food101-cifar10-method-comparison-local-56c5c70d05bce18b
Configuration: 56c5c70d05bce18b
Generated from checksummed public artifacts.

01 / START HERE

The project in plain language

Modern AI models can be adapted to new jobs without being rebuilt from scratch. The risk is that learning the new job can quietly damage abilities the model already had. This project treats that risk as something we can measure, visualize, test, and potentially control.

I adapted CLIP - a model that connects images and language - to recognize new image domains. At multiple points during learning, I measured both visible behavior (accuracy) and internal behavior (how image representations moved across every vision-transformer layer).

The central finding is not that drift is always bad. Across the completed scenarios, drift sometimes accompanied forgetting, sometimes coexisted with stable accuracy, and once accompanied positive transfer. Geometry is useful evidence, but it is not a universal substitute for testing the capability itself.

How to read the technical terms

Term	Meaning in this project
Adaptation	Teaching an existing model a new image-recognition job.
Retention	Checking whether an older capability still works afterward.
Representation	The numeric internal map the model builds for an image.
Drift	How much that internal map changes during adaptation.
CKA	A similarity score for two sets of internal representations. 1 means highly similar.
LoRA	A small trainable adapter that changes a model with far fewer parameters.
Checkpoint	A saved snapshot of the model during training.

Report map

From a course observation to a research system

The original Cornell Tech CS 5787 group project studied catastrophic forgetting while adapting CLIP from general zero-shot recognition toward Food-101. It found a visible stability-plasticity tension: the target task improved while CIFAR-10 performance declined.

The archived result was interesting but difficult to extend safely. Some saved outputs did not have their producing code, two experiment families used different schedules, and an early drift forecast failed badly. The extension therefore began by separating historical evidence from newly reproduced evidence.

> **Historical negative result: a step-8,000 forecast predicted final cosine drift of 0.6647, while the saved run ended at 0.3055 - an absolute error of 0.3592.**

A newly recovered source scaffold

A later-supplied folder named **DL Final** adds 2025 CLIP/LoRA source, tests, scripts, and two notebook templates. It adds lineage, not performance evidence: it contains no datasets, checkpoints, embeddings, results, logs, or executed notebook outputs. Static audit found seven explicit placeholders, incompatible pipeline options, missing test imports, incomplete LoRA checkpoint reconstruction, duplicate-caption false negatives, and future-information leakage in its early forecast path. It also evaluates COCO rather than the report's CIFAR-10 path, so it is not treated as the missing producer for the submitted results.

What the extension adds

Layer	Extension
Research	Multiple methods, seeds, domains, uncertainty intervals, held-out prediction protocol, and negative cases.
Diagnostics	Task metrics, calibration, fixed projections, CKA, Frechet distance, effective rank, neighborhoods, classes, layers, and cross-modal alignment.
Engineering	Configuration identity, safe checkpoints, cached immutable selections, environment capture, deterministic regeneration, schemas, and checksums.
Product	An accessible React case study with synchronized controls, Pareto comparisons, class inspection, recovery simulation, and source-level provenance.

Attribution: Sahil Mhatre, Ali Hasan, and Corey Chen completed the original 2025 group project. The rebuilt repository, corrected pipeline, new experiments, diagnostics, benchmark system, report, and interactive application are Ali Hasan's independent post-course extension.

Measure learning and forgetting at the same time

Every experiment has an adaptation domain and a retained domain. The model trains only for the adaptation objective. At saved checkpoints, both domains are evaluated, and the same retained images are passed through the frozen baseline and current model for paired geometry measurements.

SEEDS	BASE MODEL	PRIMARY ADAPTER
41 / 42 / 43	**CLIP ViT-B/32**	**294,912**
Independent model and data selections	Pinned model revision	Trainable LoRA parameters

Scientific controls

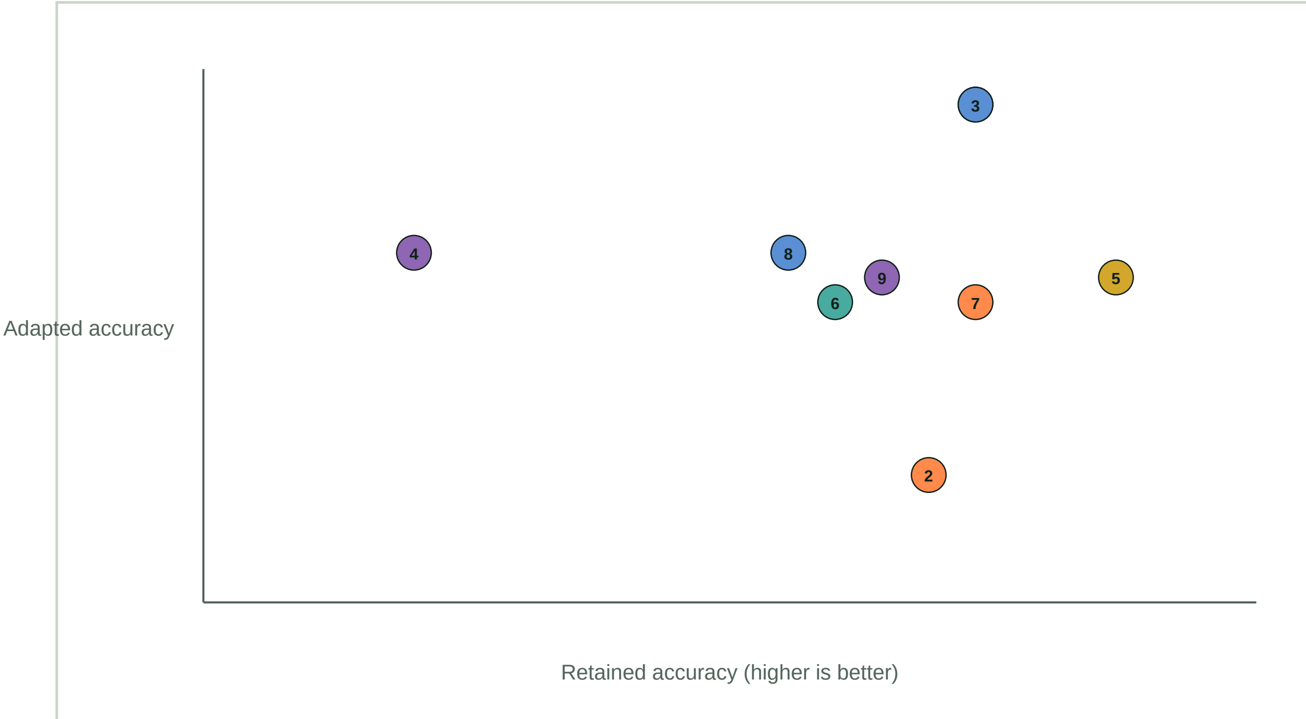
All same-class captions in a batch are positives, avoiding false negatives from repeated labels. Dimensionality reduction is fit once on baseline embeddings and reused, so plotted lines are real paired movement rather than independent t-SNE layouts. Frechet-style distance uses baseline-fixed PCA and covariance regularization. Undefined statistics, such as CKA on constant text prototypes, are published as null with an explicit definition flag.

The evidence ladder

Tier	Purpose	Publication treatment
Historical	Preserve original course artifacts and claims	Clearly labeled, not merged with reproduced results
Smoke	Test the complete CPU pipeline quickly	Engineering evidence only
Local multi-seed	Validate methods, statistics, and interactions	Preliminary, with intervals and caveats
Portfolio / research	Larger model and domain coverage	Required before strong scientific rankings

Different ways to adapt - and recover - the model

The method benchmark covers frozen probing, full-parameter tuning, initialization-aware tuning, parameter-efficient adapters, retention distillation, gradient projection, and weight-space recovery. Paper-derived methods are labeled as adapted or inspired when the local protocol is not an exact reproduction.

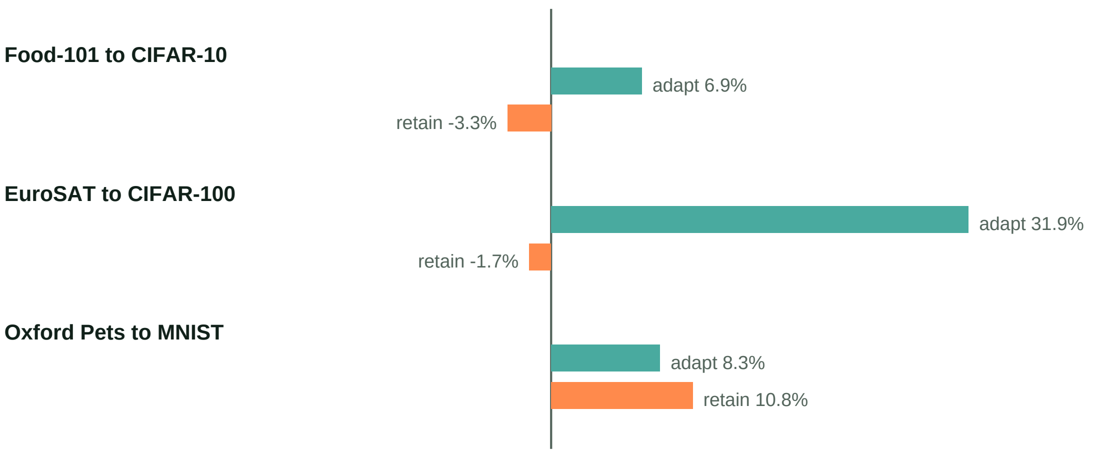


#	Method	Adapted	Retained	1 - CKA	Parameters
1	Frozen linear probe	100.0%	73.3%	0.0000	51,813
2	Full fine-tune	79.2%	72.2%	0.1011	87,901,029
3	LP-FT	100.0%	73.3%	0.0000	87,901,029
4	Zero-shot initialized FT	91.7%	60.0%	0.1778	87,901,029
5	WiSE-FT - alpha 0.5	90.3%	76.7%	0.0948	87,901,029
6	Standard LoRA	88.9%	70.0%	0.0126	294,912
7	Retention distillation	88.9%	73.3%	0.0124	294,912
8	Gradient null-space	91.7%	68.9%	0.0098	294,912
9	Selective LoRA	90.3%	71.1%	0.0086	147,456

Reading the chart: upper-right is better. The frozen probe and LP-FT reach a tiny-subset ceiling. Full fine-tuning from a random head underperforms the zero-shot adaptation baseline. WiSE-FT recovers retention from its compatible fine-tuned source. These are intervention diagnostics, not universal rankings.

Drift does not mean the same thing everywhere

The same rank-8 LoRA method was run on three adaptation and three retained domains. This tests whether the Food-101 story generalizes even before adding new model families.



Teal: adaptation change. Orange: retained-task change. Values are means across 3 seeds.

Scenario	Adaptation change	Retention change	Retained 1 - CKA	Samples / seed
Food-101 to CIFAR-10	6.9% [1.0%, 12.9%]	-3.3% [-11.6%, 4.9%]	0.0126 [0.0105, 0.0147]	24 / 30
EuroSAT to CIFAR-100	31.9% [26.0%, 37.9%]	-1.7% [-14.6%, 11.3%]	0.1558 [-0.0833, 0.3949]	24 / 40
Oxford Pets to MNIST	8.3% [-33.1%, 49.7%]	10.8% [3.7%, 18.0%]	0.0293 [0.0049, 0.0536]	12 / 40

Food-101 to CIFAR-10

Adaptation improved by 6.9 points while retention fell by 3.3 points. Mean CKA loss was small (0.0126).

EuroSAT to CIFAR-100

Adaptation improved by 31.9 points and retention fell by 1.7 points, while CKA loss was much larger (0.1558). Large drift did not imply proportionally large forgetting.

Oxford Pets to MNIST

Adaptation improved by 8.3 points and retention improved by 10.8 points while CKA loss reached 0.0293. Drift accompanied positive transfer, not forgetting.

Excluded exploratory pair

A separate Pets-to-EuroSAT artifact began at 0% retained accuracy in every seed. A floor cannot reveal additional forgetting, so the artifact is preserved as a failed validity check and excluded from this three-scenario comparison.

The most useful result is where simple stories fail

Across nine intervention means, retained CKA loss and forgetting had Pearson correlation 0.545, Spearman correlation 0.259, and a bootstrap Pearson interval [-0.790, 0.970]. The interval is too wide for a reliable method-ranking claim.

Failure case 1 - drift without forgetting

Retention distillation returned mean CIFAR-10 accuracy to baseline while internal representations still moved. Stable accuracy does not imply an unchanged model.

Failure case 2 - lower drift with worse retention

The gradient null-space baseline had lower mean CKA loss than standard LoRA but worse retained accuracy. A geometry score alone ranked these interventions incorrectly.

Failure case 3 - drift with positive transfer

Pets adaptation improved MNIST accuracy even while MNIST representations moved. Drift measures change; they do not determine whether the change helps or harms a task.

Early-warning protocol

A held-out train/validation/test evaluator was implemented with naive baselines, calibration, prediction intervals, and error metrics. On synthetic methodology data, model RMSE was 0.035 versus 0.119 for the train-mean baseline. This validates the evaluation machinery only - it is not evidence that real CLIP forgetting is predictable.

Layer and class analysis

Every CLIP vision block is measured against baseline. The application also exposes per-class accuracy, confusion counts, centroid movement, fixed-PCA sample paths, image-space drift, text-space drift, and image-text alignment. Undefined statistics remain explicit rather than being coerced into numbers.

The research pipeline is part of the result

This project is structured as a reusable system rather than a single notebook. Experiments are CLI-driven and configuration-defined. Large checkpoints and embeddings stay outside Git; compact public derivatives point to checksummed manifests.

Capability	Implementation
Identity	Run IDs and configuration hashes separate scientific settings from storage paths.
Data	Resolved repository revisions, deterministic selections, row indices, fingerprints, and revision-aware caches.
Model	Resolved model commit, trainable-parameter invariants, strategy fidelity labels, and environment versions.
Resume	Checkpointed LoRA state, optimizer state, and random-generator state; complete classification runs are idempotent.
Regeneration	Metrics and web derivatives rebuild deterministically from saved outputs without retraining.
Validation	39 Python checks, web interaction tests, TypeScript, manifest checks, responsive QA, and production budgets.
Deployment	Core charts use precomputed artifacts; no GPU or inference service is required for the public experience.

Web performance budgets

JAVASCRIPT	CORE BENCHMARK	DETAILED MICROSCOPE
~68 KB	**~199 KB**	**~846 KB**
Gzip production bundle	Initial multi-seed evidence	Loaded only on request

The visual experience was checked at mobile and wide-desktop widths. A real-browser review caught overlapping Pareto labels after the method expansion; the chart was redesigned with numbered markers and a structured legend.

What the current evidence cannot support

Limitation	Why it matters
Small local samples	Confidence intervals are wide and some tasks hit ceiling effects.
Three seeds	This is the minimum uncertainty gate, not a large statistical sample.
One foundation backbone	Nine methods and three domains still use CLIP ViT-B/32.
Short local schedules	Method outcomes may change under realistic compute and hyperparameter selection.
Reference-data asymmetry	Distillation sees retained examples; plain LoRA does not. That is a resource trade-off.
Exploratory selection	LoRA and WiSE interpolation coefficients are inspected on local evaluation data.
Association is not mechanism	Correlation between geometry and accuracy does not establish causation.

Next research gates

1. Add at least two model/control families, beginning with a modern vision-language model and a visual-only control.
2. Run larger fixed samples and realistic schedules on the three domain scenarios.
3. Add adaptive-rank LoRA and MergeTune-style recovery.
4. Train the early-warning model only after enough independent methods and scenarios exist for held-out evaluation.
5. Repeat the strongest comparisons on external GPU compute and publish compute-normalized cost.

Until those gates complete, the public experience presents measured results as preliminary protocol evidence and emphasizes the engineering, methodology, and negative findings.

How to reproduce and verify the work

Core commands

```
make test
make benchmark-local
make methods-local
PYTHONPATH=src .venv/bin/python -m driftlab aggregate-domains
cd apps/web && npm test && npm run build
```

Primary research sources

Topic	Primary source
CLIP	Radford et al. (2021), https://arxiv.org/abs/2103.00020
LoRA	Hu et al. (2021), https://arxiv.org/abs/2106.09685
CKA	Kornblith et al. (2019), https://proceedings.mlr.press/v97/kornblith19a
LP-FT	Kumar et al. (2022), https://arxiv.org/abs/2202.10054
WiSE-FT	Wortsman et al. (2022), CVPR open-access paper
ZSCL	Zheng et al. (2023), ICCV open-access paper

Artifact verification

Method comparison run: clip-food101-cifar10-method-comparison-local-56c5c70d05bce18b
Method config hash: 56c5c70d05bce18b
Method manifest SHA-256: 9ac6b060f935a0417456a872ce032be254d8df104dae8319bac0b8027fcd2d17

Domain comparison run: clip-lora-domain-comparison-local-79e5b05c661a821f
Domain config hash: 79e5b05c661a821f
Domain manifest SHA-256: b03f799b0e243940f965ac15f03e84c8852472e0bb2fbedf8dfd597357d41b12

Baseline benchmark run: clip-food101-cifar10-local-multiseed-13c5baa85a5a513c
Baseline config hash: 13c5baa85a5a513c

Historical code archive SHA-256: 957682e7d364010b8d05f452eca1e141967daf00b53eb58c2946ca3bdb5ff645

This report is generated from the same public JSON artifacts used by the portfolio application. If a source manifest checksum changes, the production web build rejects the mismatch.